Elastic Bending of a Cantilever Beam: Experimental Determination of Young's modulus

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Abstract

Cantilever beams are structural elements that are fixed at one end and free at the other, and are widely used in engineering applications where loads cause bending. This experiment investigated the elastic deflection of a metal cantilever beam in order to determine the Young's modulus of the beam material. A rectangular metal bar was clamped at one end, and known masses were suspended from the free end. The resulting vertical deflection was measured relative to the unloaded position.

Two experiments were performed. First, the effect of cantilever length on deflection was examined under a constant applied load. Second, the effect of applied load on deflection was studied for a fixed cantilever length. In both cases, linear relationships were observed between deflection and applied load, and between deflection and the cube of the cantilever length, consistent with elastic beam theory.

Young's modulus was calculated using graphical methods, giving values in the range of 1.3×10^{11} Pa *to* 1.7×10^{11} Pa*, which are of the correct order of magnitude for a steel alloy. The load-deflection method produced results closer to reference values, due to reduced sensitivity to experimental uncertainty compared to the length-deflection method. Differences from reference values were attributed to measurement uncertainty and experimental limitations.*

1 Introduction

1.1 Elastic behaviour of materials

When a force is applied to a solid material, it undergoes deformation. If this deformation is small and reverses once the load is removed, the material is said to behave elastically. Within these elastic limits, stress is proportional to strain, a relationship known as Hooke's law (Hibbeler 2017). Young's modulus, E, is the constant in this proportional relationship, and provides a measure of the material's stiffness. Materials with a high Young's modulus deform less under the same stress than materials with a lower Young's modulus. Young's modulus is therefore essential in engineering, as it allows engineers to predict how materials will deform under load (Hibbeler 2017; Young and Budynas 2002).

1.2 Cantilever beams in engineering

A cantilever beam is a structural element that is fixed at one end and free at the other. This configuration is widely used in engineering applications such as balconies, crane arms, and aircraft wings, where material stiffness directly controls deflection (Patel et al. 2015). Cantilevers are also used at smaller scales, including in sensors and electromechanical systems. Excessive deflection can lead to plastic deformation and structural failure, making accurate prediction of bending behaviour crucial.

According to Euler-Bernoulli beam theory, the deflection, δ, of a cantilever beam with a rectangular cross-section depends on the load, W, the beam length, L, the cross-sectional width, a, the cross-sectional height, b, and the Young's modulus of the material, E (Young and Budynas 2002). For a beam of constant material and cross-section, the deflection is directly proportional to the applied load and the cube of the beam length.

$$\delta = \frac{4WL^3}{Eab^3} \tag{1}$$

1.3 Significance of the experiment

Determining Young's modulus experimentally demonstrates a link between the theoretical behaviour and real material behaviour. Although reference values of Young's modulus are widely available, determining it through experimental data helps us demonstrate how these values arise, and how experimental uncertainty affects the final result. Furthermore, cantilever experiments require relatively simple equipment but demonstrate fundamental principles (Turvey 1989).

Similar cantilever-based methods have been shown to produce reliable estimates of Young's modulus in undergraduate laboratory settings, provided deflections are sufficiently large to minimise relative measurement uncertainty (Turvey 1989).

2 Aim & Objectives

The aim of this experiment is to investigate the elastic bending behaviour of a cantilever beam and to determine the Young's modulus of the beam material from experimental deflection measurements. The objectives of the experiment were to:

- Measure the vertical deflection of a cantilever beam as a function of its length under a constant applied load.
- Measure the vertical deflection of a constant length cantilever beam as a function of applied load.
- Calculate Young's modulus using Euler-Bernoulli beam theory and compare results obtained from both experimental methods with reference values.
- Evaluate causes of experimental uncertainty and identify improvements for future experiments.

3 Experimental Methodology

3.1 Apparatus and materials

The experiment was performed using a rectangular metal bar acting as a cantilever beam. The beam was rigidly clamped at one end and attached to a laboratory workbench. The beam had a total length of 1.00 m, a mass of 486 g, a width of 13.24 mm, and a thickness of 5.18 mm, with a uniform cross-sectional geometry.

A set of masses and a load hanger were used to apply loads to the beam. A ruler was used to measure the vertical height of the beam's free end relative to the laboratory floor. All measurements were recorded manually.

3.2 Experiment setup

The cantilever beam was mounted horizontally with one end rigidly clamped, leaving the remainder of the beam free to bend under load. The unclamped length of the beam was initially measured to be 0.947 m. An initial reference height was recorded by measuring the vertical distance from the floor to the free end of the unloaded beam. This reference height was used to calculate deflection for subsequent measurements.

Loads were applied by suspending masses from the beam. Care was taken to ensure that the load acted perpendicular to the beam, that the load point did not move during loading, and that the beam was not twisted during loading. The experimental setup is photographed in Figure 1.

Similar experimental setups using simple mechanical measurement techniques have been reported in undergraduate laboratory studies of cantilever bending (Turvey 1989; Biswas 2022).



Figure 1: *Measuring deflection of cantilever beam*

3.3 Experiment procedure

Effect of cantilever length A constant mass of 1.219 kg (11.96 N) was suspended from the free end of the beam. The effective cantilever length was varied by repositioning the beam within the clamp. For each length, the vertical height of the free end was measured and the deflection was calculated relative to the unloaded reference height.

Effect of applied load The effective cantilever length was fixed at 0.706 m. Masses ranging from 0.218 kg to 2.713 kg were applied incrementally at the free end. For each load, the vertical height was measured and the corresponding deflection calculated.

3.4 Variables

The independent variables were the applied load and the effective cantilever length. The dependent variable was the vertical deflection of the beam. Controlled variables included the beam geometry, material, clamping arrangement, and measurement method.

3.5 Measurement accuracy and sources of uncertainty

Several sources of uncertainty were present in the experiment. The primary source of error arose from measuring the beam's vertical deflection using a ruler, which limited the precision of the height readings. Although the ruler was graduated in millimetres, accurate measurements required it to be held perpendicular to the floor and aligned with the direction of deflection. Small misalignments, together with

parallax when reading the scale, could therefore affect the recorded values. To reduce ambiguity when identifying the beam's vertical position, a straight edge was placed on top of the beam to provide a clear horizontal reference line against the ruler; however, small errors remained due to alignment and the parallax effect.

Further uncertainty resulted from neglecting the bending of the beam under its own weight during initial calculations. Although this effect is small compared to the applied loads, it can influence the accuracy of the calculated Young's modulus. These sources of uncertainty are considered further in the Results and Discussion section.

Deflection-based methods are particularly sensitive to measurement uncertainty when deflections are small (Farsi et al. 2017; Turvey 1989).

3.6 Safety considerations

The experiment involved suspended masses, which posed a risk if they were dropped accidentally. Care was taken to ensure that the masses were securely attached before release and that hands and feet were kept clear of the experiment area. The beam clamp was checked after each loading to ensure that it was secure and stable throughout the experiment.

4 Theoretical Analysis

4.1 Elastic bending and Young's modulus

When a load is applied to a beam, it causes the beam to deform. If the applied load is sufficiently small, the beam will return to its original shape when the load is removed and the deformation is said to be elastic. In this region of elastic behaviour, amount of deflection is characterised by Young's modulus, E, which relates stress to strain and provides a measure of material stiffness (Hibbeler 2017; Young and Budynas 2002).

The magnitude of deflection of a loaded beam depends on the applied load, the geometry of the beam, and the elastic properties of the material. Measuring beam deflection with a known geometry and load therefore provides an experimental method for estimating the Young's modulus of the beam material.

4.2 Deflection of a cantilever beam under a point load

In this experiment, a cantilever beam was fixed at one end and loaded at the free end with a known weight. The vertical deflection of the free end, δ, was measured relative to the initial unloaded position (ignoring the load of the beam self-weight).

For a rectangular cantilever beam of length L, width a, and thickness b, subjected to a point load W at the free end, the vertical deflection is given by

$$\delta = \frac{4WL^3}{Eab^3} \tag{2}$$

where δ is the vertical deflection (m), W is the applied load (N), L is the effective cantilever length (m), E is Young's modulus of the beam material (Pa), a is the width of the beam (m), and b is the thickness of the beam (m).

This equation is derived from Euler-Bernoulli beam theory for small deflections of cantilever beams (Hibbeler 2017; Young and Budynas 2002), and is the basis for all other equations in this experiment.

4.3 Effect of cantilever length on deflection

In the first part of the experiment, the applied load was kept constant while the effective length of the beam was varied. From Equation 2, the deflection can be expressed as

$$\delta \propto L^3 \tag{3}$$

This relationship predicts that deflection will increase rapidly with beam length (being proportional to the cube of the beam length). To test this, recorded values of deflection were plotted against L^3 (shown in Figure 2), expecting a linear relationship between δ and L^3 which would indicate agreement between the theoretical model and our experimental results.

Rearranging Equation 2 gives an equation for Young's modulus in terms of the gradient of the graph:

$$E = \frac{4W}{ab^3 s} \tag{4}$$

where s is the gradient of the deflection versus L^3 plot. This method was used to estimate Young's modulus from the experimental data.

4.4 Effect of applied load on deflection

In the second part of the experiment, the cantilever beam length was kept constant, while the load applied to the end was varied. From Equation 2, deflection is directly proportional to the applied load:

$$\delta \propto W \tag{5}$$

This predicts a linear relationship between deflection and load. Experimental values of deflection were plotted against applied load (shown in Figure 3), and the resulting linear regression best-fit line was used to determine Young's modulus.

Rearranging Equation 2 for this case gives

$$E = \frac{4L^3}{ab^3} \cdot \frac{W}{\delta} \tag{6}$$

or, using the gradient s of the deflection versus load graph,

$$E = \frac{4L^3}{ab^3\, s} \tag{7}$$

This approach provides an alternative estimate of Young's modulus, allowing comparison between the two methods used in this experiment.

4.5 Correction for the beam's self-weight

In addition to the applied load, the beam's own weight contributes to deflection. To account for this effect, the applied load can be modified by adding a term representing the beam's weight:

$$W_{\text{eff}} = W + cmg \tag{8}$$

where m is the mass of the beam (kg), g is the acceleration due to gravity ($\mathrm{m\,s^{-2}}$), and c is a dimensionless constant that accounts for how the distributed weight of the beam contributes to deflection at the free end.

Substituting Equation 8 into Equation 2 gives

$$\delta = \frac{4L^3}{Eab^3}\,(W + cmg) \tag{9}$$

This equation predicts a straight-line relationship between deflection and applied load with a non-zero intercept. The intercept can be used to estimate the value of c.

$$c = \frac{\delta_0}{s \cdot mg} \tag{10}$$

where δ_0 is the Y-intercept and s is the slope. Accounting for the beam's self-weight as a distributed load is standard practice in more accurate cantilever analyses (Young and Budynas 2002; T. Beléndez, Neipp, and A. Beléndez 2003).

4.6 Validity of the theoretical model

The theoretical basis for this experiment assumes that the beam behaves elastically, deflections are small compared to the beam length, the beam has a uniform rectangular cross-section and material, and the clamp provides a perfectly rigid point for the beam to bend around. These assumptions are consistent with standard examples in mechanical science textbooks (Hibbeler 2017), and are commonly used in undergraduate experiments (Turvey 1989; T. Beléndez, Neipp, and A. Beléndez 2002).

These expressions can be derived more rigorously using Euler-Bernoulli beam theory, however this experiment relies only on the resulting deflection formula and proportional relationships, rather than the full differential equation forms. More advanced forms are discussed in literature, but are beyond the scope of this experiment (T. Beléndez, Neipp, and A. Beléndez 2002).

5 Results & Discussion

Table 1: *Effect of cantilever length, L, on deflection, δ, under constant load of 11.96 N*

Length (m)	Height (m)	L^3 ($\mathrm{m^3}$)	δ (m)
0.344	0.945	0.041	0.002
0.494	0.929	0.121	0.018
0.602	0.909	0.218	0.038
0.739	0.866	0.404	0.081
0.928	0.795	0.799	0.152

Table 2: *Effect of applied load on deflection, δ, for constant cantilever length of 0.706 m*

Load (N)	Height (m)	δ (m)
2.14	0.927	0.033
7.01	0.906	0.054
11.95	0.882	0.078
16.81	0.860	0.100
21.74	0.837	0.123
26.61	0.816	0.144

5.1 Effect of cantilever length on deflection

In the first experiment, the applied load was kept constant at 11.96 N while the cantilever beam length was varied by repositioning the beam within the clamp. The deflection was calculated using an unloaded reference height of 93.0 cm. The measured heights and calculated deflections are summarised in Table 1.

The beam can be viewed as a lever about the fixed clamp, where increasing the beam length increases the moment of the applied load and hence the bending moment at the clamp. This larger bending moment leads to greater curvature along the beam, resulting in an increase in deflection with length, consistent with the predicted $\delta \propto L^3$ relationship.

When the deflection δ was plotted against L^3, as shown in Figure 2 , a linear relationship was observed. This behaviour is consistent with the theoretical prediction that deflection is proportional to the

cube of the cantilever length, as given by Equation 3. From the gradient of the δ versus L^3 graph, Young's modulus was calculated using Equation 4 and the slope of the graph, giving a value of approximately 1.289×10^{11} Pa.

Although this value is lower than typical reference values for steel, approximately 2.0×10^{11} Pa (Hibbeler 2017), it is of the correct order of magnitude. Furthermore, the exact alloy composition of the cantilever beam used is unknown. The discrepancy is likely due to experimental uncertainty, particularly in measuring effective length accurately. Because deflection depends on L^3, small errors in length measurement produce relatively large errors in the calculated Young's modulus.

5.2 Effect of applied load on deflection

In the second experiment, the effective cantilever beam length was fixed at 0.706 m and the load applied to the end was varied. The measured heights and calculated deflections are shown in Table 2.

The results in Table 2 show that deflection increases approximately linearly with applied load. A plot of deflection δ against load W produced a straight-line relationship, consistent with the theoretical prediction given by Equation 5.

Using the gradient of the linear regression of the data in Figure 3 and Equation 7, Young's modulus was calculated as approximately 1.663×10^{11} Pa. This value is closer to commonly cited reference values for steel than the value obtained from the length-deflection experiment, suggesting that this method may be less sensitive to experimental uncertainty.

5.3 Comparison of Young's modulus values

The two experimental methods produced different estimates of Young's modulus. The length-deflection method yielded $E \approx 1.289 \times 10^{11}$ Pa, while the load-deflection method yielded $E \approx 1.663 \times 10^{11}$ Pa. The mean value of these two estimates was 1.48×10^{11} Pa.

The difference between the two values can be explained by the different sensitivities of the methods to experimental uncertainty. In the length-deflection experiment, uncertainty in measuring the effective cantilever length is significantly increased due to the cubic relationship to L. In contrast, the load-deflection experiment depends linearly on the applied load, reducing the impact of uncertainty in load measurement.

5.4 Influence of beam self-weight

The beam self-weight contributes to bending, even when there is no applied load. When the own weight correction described in Equation 9 was applied, the deflection versus load graph shows a non-zero intercept of 0.0229. From this intercept, the constant c was estimated to be approximately 0.37 according to Equation 10.

5.5 Experimental uncertainty and limitations

The primary source of uncertainty in this experiment was the measurement of deflection using a ruler. The deflections were relatively small, making the results sensitive to parallax error, and the necessity of holding the ruler perfectly parallel to the direction of load and perpendicular to the beam. The deflection was read from a ruler with 1 mm graduations, but in practice the reading was limited by parallax and alignment. A conservative reading uncertainty of ± 5 mm was assumed for δ.

Additional uncertainty arose from accurately measuring the effective cantilever length.

Despite these limitations, the experimental results show clear linear trends that agree with theoretical predictions. Similar levels of uncertainty and deviation from ideal behaviour have been reported in comparable cantilever beam experiments at an undergraduate level (Turvey 1989; T. Beléndez, Neipp, and A. Beléndez 2002).

6 Conclusions

This experiment investigated the elastic bending behaviour of a cantilever beam and used deflection measurements to estimate the Young's modulus of the beam material. Deflection was measured as a function of cantilever length under a constant load and as a function of applied load for a fixed cantilever length. In both cases, the experimental results showed linear trends consistent with theoretical predictions for elastic beam bending.

Young's modulus was estimated as 1.29×10^{11} Pa using the length-deflection method and 1.66×10^{11} Pa using the load-deflection method, giving a mean value of 1.48×10^{11} Pa. This value is of the correct order of magnitude for a steel alloy. Differences between the two estimates were attributed primarily to measurement uncertainty, particularly in determining deflection and effective cantilever length, as well as the influence of the beam's self-weight.

Despite these limitations, the experiment successfully demonstrated the relationship between beam

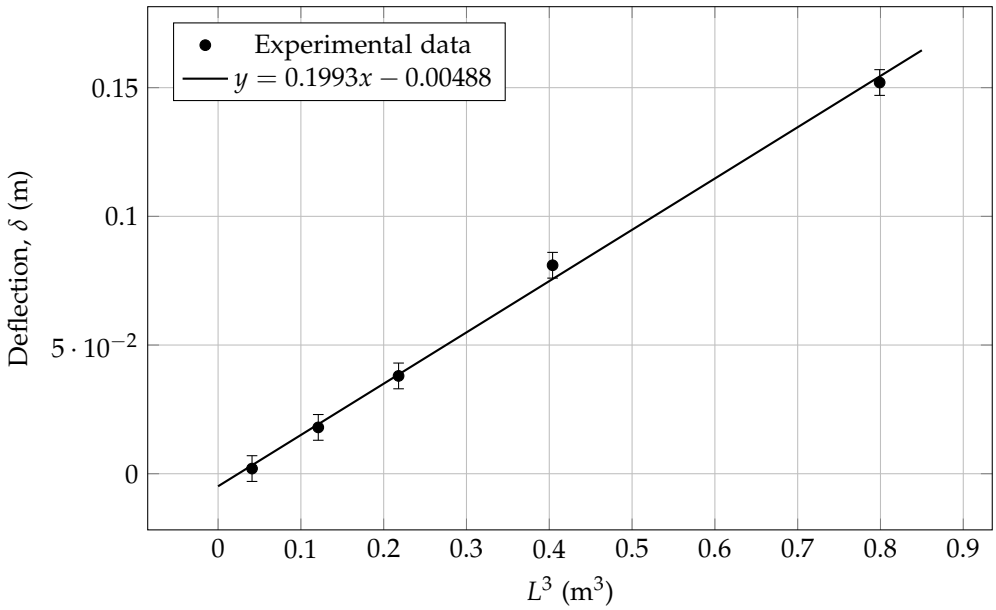


Figure 2: *Deflection of the cantilever beam as a function of L^3 under constant applied load of 11.96 N, with linear best-fit trendline*

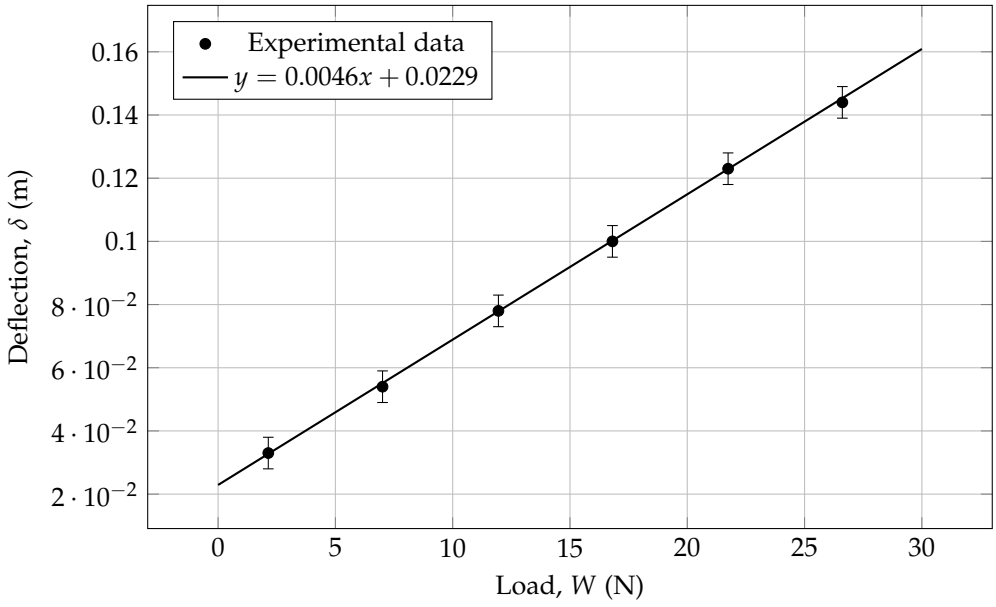


Figure 3: *Deflection of the cantilever beam as a function of applied load with linear best-fit trendline*

geometry, applied load, and elastic deflection, highlighting the practical importance of Young's modulus in engineering design and structural analysis.

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